Natural Gas Well Feasibility Analysis Report

Well: PEGASUS (DEVONIAN) Well No.: 70279125

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Project Type: Self-Initiated Project

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1. Executive Summary

This report presents a comprehensive feasibility analysis for the PEGASUS (DEVONIAN) Well, No. 70279125, utilizing a multi-faceted approach encompassing Decline Curve Analysis (DCA), Nodal Analysis, and a detailed economic evaluation, including a robust scenario analysis. The primary objective was to assess the well's long-term production potential and economic viability under current and varied market conditions.

The Decline Curve Analysis, based on 84 months of historical gas and condensate production, projects an Estimated Ultimate Recovery (EUR) for gas and condensate over a 444-month production life. The economic evaluation reveals a **Base Case Net Present Value (NPV) of -$5,345,626.11**, indicating that the project is currently not economically viable under the specified assumptions.

Furthermore, Nodal Analysis identified the well's optimal operating point at **4166.67 MCF/month (gas flow rate) and 2583.33 psi (bottom-hole flowing pressure)**, but also highlighted the critical issue of **liquid loading below 3125 MCF/month**, which significantly impedes sustained production at lower flow rates.

Scenario analysis, varying gas price, operating expenditure (OPEX), and discount rates, showed that even under optimistic conditions, the project's NPV remains negative, reinforcing its current inviability and high sensitivity to key economic parameters. Practical operational recommendations, such as the implementation of plunger lift systems or velocity strings, are proposed to mitigate liquid loading and enhance production efficiency.

This methodology (DCA, NPV evaluation, nodal analysis) is directly applicable to Mozambique’s onshore gas fields (e.g., Temane/Inhassoro), where similar production optimization and economic evaluation challenges are encountered.

2. Introduction

The PEGASUS (DEVONIAN) Well, No. 70279125, represents a significant asset requiring thorough technical and economic assessment to determine its long-term viability. This report provides a detailed analysis, integrating various petroleum engineering and economic principles. The study aims to forecast future hydrocarbon production, evaluate project profitability, identify operational challenges, and assess sensitivity to market fluctuations. The findings will serve as a critical foundation for strategic decision-making regarding the well's future development and operational strategies.

3. Methodology

This section outlines the analytical methods employed to conduct the feasibility study of the PEGASUS Well.

3.1. Data Acquisition and Preparation

Historical production data, spanning 84 months for both gas and condensate, was acquired and prepared. This data included monthly gas production (MCF), condensate production (BBL), and corresponding time series. These historical trends formed the basis for forecasting future production.

3.2. Decline Curve Analysis (DCA)

Decline Curve Analysis was performed on the historical production data to forecast future gas and condensate production rates. An **exponential decline model** was utilized, identified as the best fit for the observed production trends. The key parameters determined from the historical data are:

- Initial Production Rate (qi) for Gas: **66,087 MCF/month**
- Initial Decline Rate (Di) for Gas: **0.018 1/month**
- Condensate Factor: **77.69 MCF/BBL** (derived as the average historical gas-to-condensate ratio)

These parameters were used to project production for a total economic life of 444 months (37 years).

3.3. Economic Evaluation

A comprehensive monthly cash flow model was developed to assess the project's economic viability. The model incorporates:

- **Revenue Generation:** Calculated from forecasted gas and condensate production multiplied by their respective prices.

- **Operating Expenditure (OPEX):** Assumed as a fixed monthly cost.
- **Capital Expenditure (CAPEX):** Represents the initial investment.

The Net Present Value (NPV) was calculated by discounting the cumulative cash flow over the project's life using a defined discount rate.

3.4. Nodal Analysis

Nodal Analysis was conducted to evaluate the well's performance by identifying the optimal operating point and potential flow limitations. This involved plotting:

- **Inflow Performance Relationship (IPR):** Represents the reservoir's ability to deliver fluids to the wellbore.
- **Vertical Lift Performance (VLP):** Represents the wellbore's ability to lift fluids to the surface.

The intersection of the IPR and VLP curves indicates the well's optimal operating point. The analysis also focused on identifying the critical flow rate to assess the risk of liquid loading.

3.5. Scenario Analysis

To evaluate the project's sensitivity to market fluctuations and operational costs, a scenario analysis was performed. Three distinct cases were considered:

- **Best Case (Optimistic):** Higher gas prices, lower OPEX, and a lower discount rate.
- **Base Case:** Standard assumptions.
- **Worst Case (Pessimistic):** Lower gas prices, higher OPEX, and a higher discount rate.

The NPV was calculated for each scenario to understand the range of potential outcomes.

3.6. Economic Assumptions Table

The following table summarizes the key economic parameters used in the Base Case evaluation. These values were systematically varied for the scenario analysis.

Parameter	Value	Unit	Source/Assumption
Gas Price (Base)	3.00	$/MCF	Market Reference (EIA) / Assumed
Oil Price (Base)	70.00	$/BBL	Market Reference / Assumed
Monthly OPEX (Base)	3,000.00	$/month	Assumed Industry Average
Monthly Discount Rate (Base)	0.83%	% / month	Equivalent to 10% Annual Discount
CAPEX	-15,000,000.00	$	Initial Investment (Assumed)

4. Results

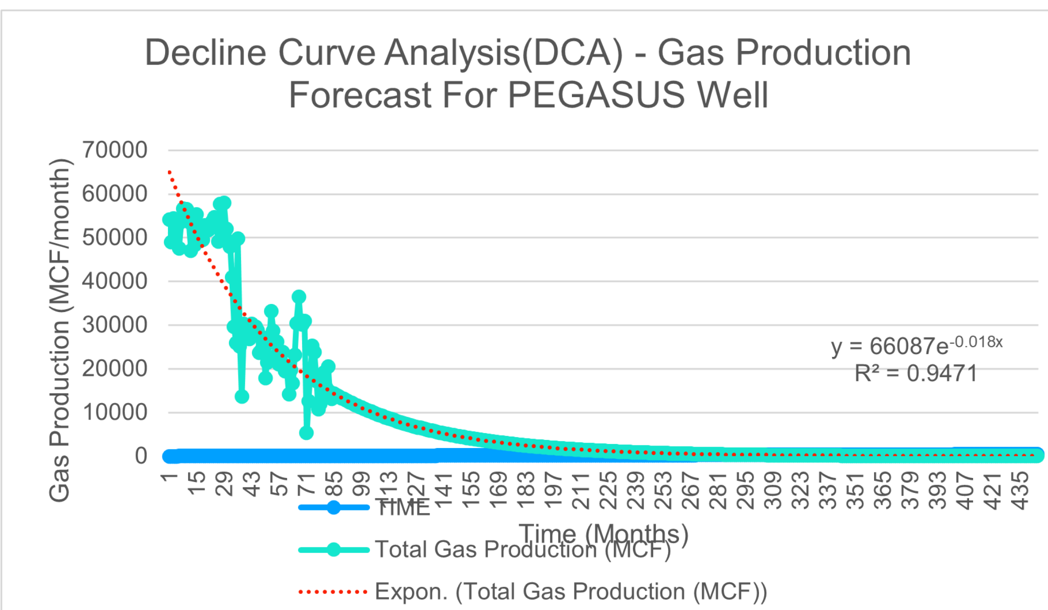
This section presents the key findings from the Decline Curve Analysis, Economic Evaluation, and Nodal Analysis.

4.1. Decline Curve Analysis Results

The exponential decline model accurately fits the historical production data, as evidenced by an R^2 value of 0.9471. The projected production curve indicates a continued decline over the well's economic life.

- **Estimated Ultimate Recovery (EUR) for Gas: 3,710,384.63 MCF**
- **Estimated Ultimate Recovery (EUR) for Condensate: 50,388.44 BBL**

Figure 1: Decline Curve Analysis - Gas Production Forecast for PEGASUS Well. The graph illustrates the historical gas production data (dots) and the fitted exponential decline curve, extending the forecast over the well's life.

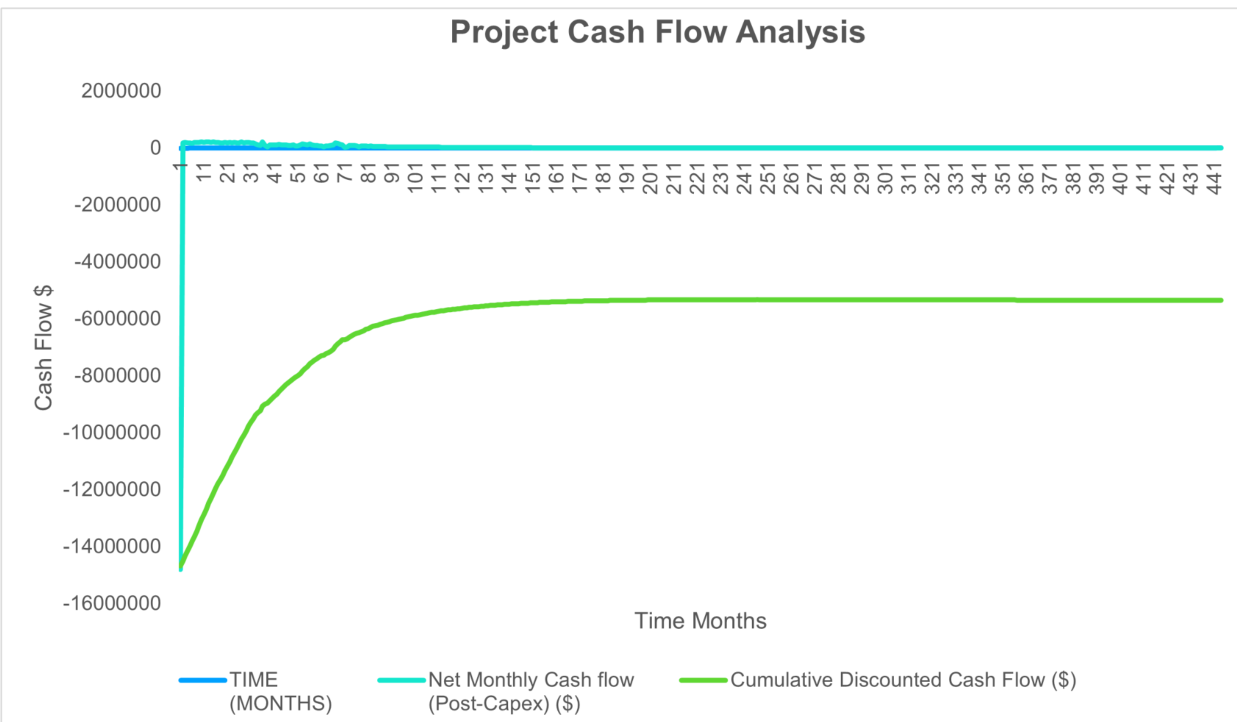


4.2. Economic Evaluation Results

The economic evaluation, based on the forecasted production and specified economic assumptions, yielded the following results for the Base Case:

- **Net Present Value (NPV): -$5,345,626.11**
- **Payback Period:** The cumulative discounted cash flow remains negative throughout the project's life, indicating that the project does not achieve payback under the current economic assumptions.

Figure 2: Project Cash Flow Analysis. This graph shows the monthly cash flow post-CAPEX and the cumulative discounted cash flow over the project's life. The negative NPV indicates the project's economic inviability.

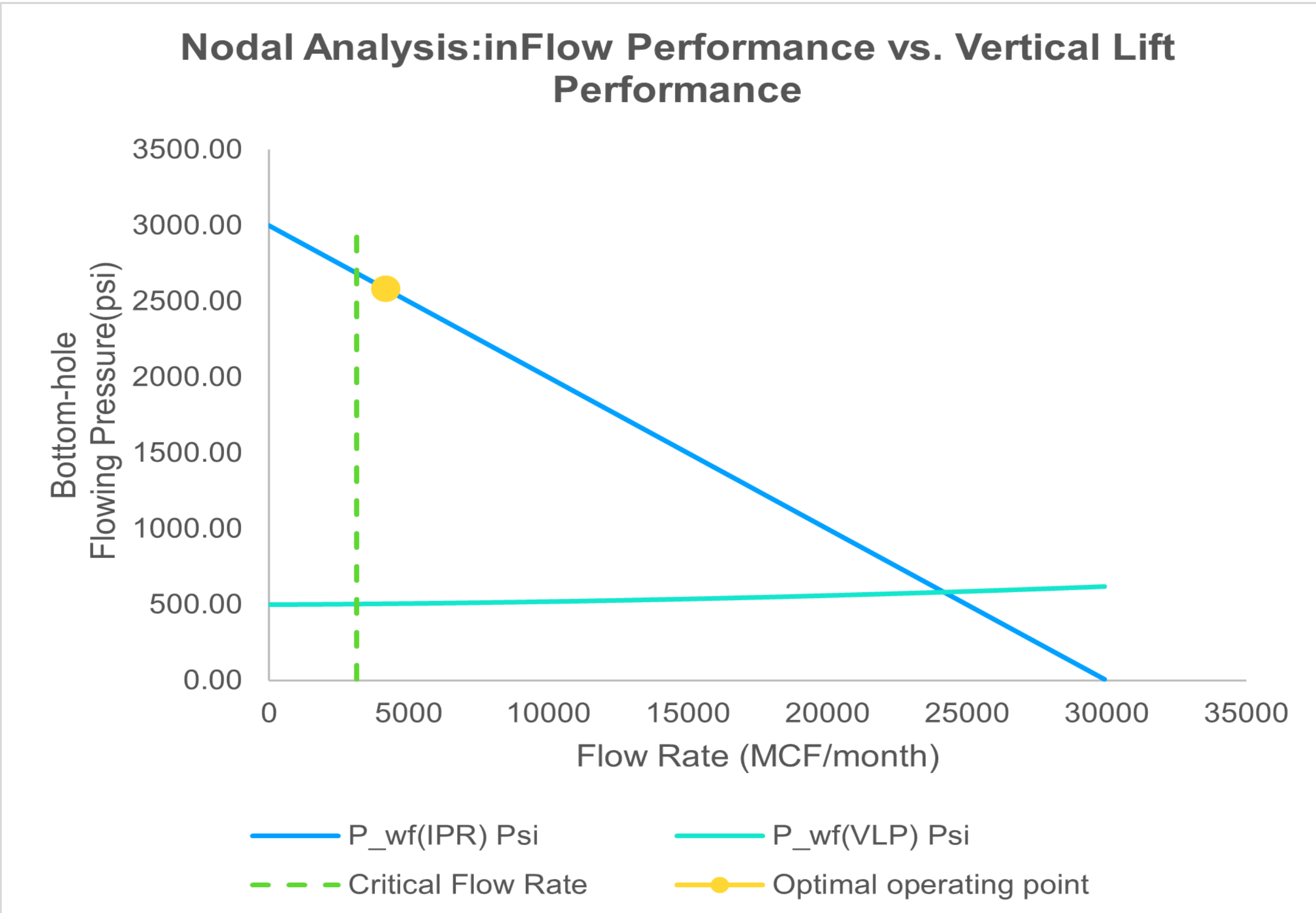


4.3. Nodal Analysis Results

The Nodal Analysis highlights the dynamic interplay between the reservoir's deliverability (IPR) and the wellbore's lifting capability (VLP).

- **Optimal Operating Point:** The intersection of the IPR and VLP curves defines the most efficient operating conditions for the well.
  - **Optimal Flow Rate (Q): 4166.67 MCF/month**
  - **Optimal Bottom-hole Flowing Pressure (Pwf): 2583.33 psi**
- **Liquid Loading:** The analysis identified a critical issue of liquid loading, which occurs when the gas velocity is insufficient to carry liquids out of the wellbore, leading to fluid accumulation and increased bottom-hole pressure.
  - **Critical Flow Rate (Q_critical): 3125 MCF/month**
  - Below this rate, the well is highly susceptible to liquid loading, potentially leading to production decline or cessation.

Figure 3: Nodal Analysis: Inflow Performance vs. Vertical Lift Performance. The graph displays the IPR and VLP curves, their optimal intersection point, and the critical flow rate marking the onset of liquid loading.



5. Scenario Analysis

The sensitivity analysis demonstrates the project's vulnerability to changes in key economic parameters. The NPV was calculated for three scenarios:

Case	**Gas Price ($/MCF)**	**OPEX ($/month)**	**Discount Rate (%)**	**NPV ($)**
Best	4.50	2,400	0.80%	**-1,335,180.21**
Base	3.00	3,000	0.83%	-5,345,626.11
Worst	2.00	3,600	1.20%	**-8,729,472.01**

6. Operational Recommendations

Based on the comprehensive analysis, the following recommendations are put forth:

- **Economic Re-evaluation:** Given the consistent negative NPV across all scenarios, it is recommended not to proceed with the project under the current economic and operational assumptions. A re-evaluation should be considered only if there are significant and sustained increases in gas prices or substantial reductions in CAPEX and OPEX.
- **Liquid Loading Mitigation:** The identified risk of liquid loading at flow rates below 3125 MCF/month is a critical operational challenge. To mitigate this and sustain long-term gas production efficiency, it is recommended to investigate the feasibility of implementing artificial lift solutions. These may include:
    - **Plunger Lift Systems:** Cost-effective solution for intermittent liquid removal.
    - **Foam Injection or Surfactants:** To reduce liquid holdup and lower the density of the fluid column.
    - **Velocity String Installation:** To increase gas velocity and reduce the critical flow rate required to lift liquids.
- **Further Studies:** Detailed engineering studies are required to assess the technical and economic viability of these remedial actions, including a cost-benefit analysis for each proposed solution.

7. Conclusion

The feasibility analysis of the PEGASUS (DEVONIAN) Well, No. 70279125, reveals that the project, despite its production potential, is not economically viable under the current or even optimistic market assumptions, as indicated by a consistently negative Net Present Value. The primary operational challenge identified is liquid loading, which will significantly impair long-term production. Strategic interventions aimed at mitigating liquid loading are essential for enhancing well performance, but even with such improvements, a re-assessment of overall economic viability is crucial. The insights derived from this analysis underscore the importance of comprehensive engineering and economic modeling in guiding upstream investment decisions.

8. Lessons Learned

This project provided invaluable insights into the integrated approach required for petroleum well feasibility studies. Key lessons include:

- **Interdependence of Technical and Economic Factors:** The project's economic viability is directly tied to accurate production forecasting and the ability to manage operational challenges like liquid loading.
- **Importance of Sensitivity Analysis:** Varying key parameters (prices, costs, discount rates) demonstrated the project's high sensitivity to these factors, highlighting the need for robust risk assessment in investment decisions. Even seemingly small changes can have a substantial impact on profitability.
- **Practical Application of Engineering Principles:** Applying Decline Curve Analysis and Nodal Analysis in a real-world context solidified the understanding of well performance and production optimization strategies.
- **The Value of Iterative Modeling:** The process of refining the model, correcting errors (e.g., in qi value), and ensuring consistency across sheets underscored the importance of meticulous attention to detail in spreadsheet modeling.
- **Communication of Complex Results:** The exercise of structuring and presenting technical and economic findings in a clear, professional report format is crucial for effective communication to stakeholders.